SPACE X · COPY INTO NOTEBOOK

Concept cards for the word bank · S1–S3 (v1)

32 core terms · two cards per notebook page · **16 pages (8 spreads), numbered C1–C16** · 309 handwritten lines. Every page that follows is one notebook spread at full size: copy it line for line.

The five symbols (write this key at the top of C1's facing page or on your abbreviation page)

Write	Means	What that line does for you
[NAME] = …	official name = also called	Box the name in pencil. Say it out loud before you write the rest.
=	meaning	The dictionary line. What the word points at.
WHY	why it exists	The problem it solves. This is the line that makes the word stick.
#	real numbers	One example you can check. No number = you can't test yourself.
≠	don't confuse with	The look-alike, and in brackets how to tell them apart.
→	connects to	Formulas and terms it plugs into. Your next sentence in a conversation.

Pencil only. Nothing here needs color. The tiny gray numbers 1–22 down the left edge are your ruled lines; don't copy them. The dashed line is one skipped line between the two cards.

How to copy so it actually sticks

1. **One spread per sitting** (4 cards, about 40 lines, 20–25 minutes). Eight sittings. Don't do all 16 pages in one go; by page 5 you would be copying shapes, not ideas.
2. **Order:** C3–C7 first (scene 3, the orbital elements: your biggest gap and the heart of the 7-minute talk). Then C11–C15 (fuel bill + flybys). Then C1–C2, C8–C10, C16.
3. **Per card:** read the whole card. Say the WHY line out loud. Then write the card. Writing first and reading later is the wrong way round.
4. **Close the loop:** when a spread is done, cover everything except the four boxed names and say each WHY from memory. Miss one twice → it becomes a question-bank card.
5. **Link the two banks:** in the one-line word bank, write "→ C5" (the card's page) at the end of that term's line. Index of all 32 is on the last page of this PDF.

Where the 16 pages go, and what it costs the rest of the notebook, is on the last page. Short version: they start 44 pages from the end, right in front of the talk cards. Nothing you have already written moves.

WORD BANK · CONCEPT CARDS · scene 1 **C1**

CENTRIPETAL FORCE = center-seeking force scene 1

= inward pull that bends a straight path into a circle

WHY nothing circles on its own. For a sat the pull IS gravity, so no engine is needed

\# ISS: 8.7 m/s² inward at 7.67 km/s = what a 6,771 km circle needs

≠ centrifugal 'force' (your inertia)

→ v = √(μ/r): gravity = needed pull

STATION-KEEPING = orbit maintenance scene 1

= small burns that cancel drift

WHY drag, Moon, Sun, lumpy Earth nudge every orbit. Fixing it burns fuel = what ends a sat's life

\# GEO ~50 m/s per yr · ISS reboost a few m/s monthly

≠ 'holding it up' (nothing does)

→ reboost · perturbation · graveyard

WORD BANK · CONCEPT CARDS · scene 2 **C2**

GEOSTATIONARY ORBIT (GEO) = Clarke orbit scene 2

= 1-day lap, circular, over equator: stays put in the sky

WHY a fixed dish can point at it forever. Needs T = 1 Earth spin: true at one height only

\# alt 35,786 km · r 42,164 km · 3.07 km/s · 23 h 56 m

≠ geosynchronous (tilted: figure-8)

→ T = 2π√(a³/μ) · GTO · station-keeping

CONSTELLATION scene 2

= many sats in planned orbits working as one system

WHY one sat = one place, and low sats move fast. Need enough planes so one is always overhead

\# GPS 24+ in 6 planes · Iridium 66 · Starlink 1000s

≠ sats that just share an orbit

→ footprint · revisit time · planes

WORD BANK · CONCEPT CARDS · scene 3 **C3**

CLASSICAL ORBITAL ELEMENTS = Keplerian scene 3

elements (COEs)

= a e i Ω ω ν: six numbers that fully describe an orbit

WHY position + velocity change every second. Describe the PATH instead and five barely change

ISS: a 6,790 km · e 0.0005 · i 51.6° · ν laps in 92 min

≠ lat/long (a ground point, not a path)

→ vis-viva and T use a · burn rules

SEMI-MAJOR AXIS (a) = size of the orbit scene 3

= half the long width of the ellipse

WHY one number for 'how big'. Lap time depends ONLY on a

ISS 6,790 km → 92 min · GPS 26,560 → 12 h · GEO 42,164 → 24 h

≠ altitude (height above ground)

→ $T = 2\pi\sqrt{(a^3/\mu)}$ · $a = (r_1+r_2)/2$

WORD BANK · CONCEPT CARDS · scene 3 **C4**

ECCENTRICITY (e) = shape · squash scene 3

= how far the ellipse is from a circle

WHY an ellipse needs one 'how squashed' number: 0 = circle, 1 = barely open

ISS 0.0005 · GTO 0.73 · escape > 1

≠ inclination (e squash, i tilt)

→ $r_p = a(1-e)$ · $r_a = a(1+e)$

INCLINATION (i) = tilt scene 3

= angle between orbit plane and equator

WHY sets what you fly over: never past latitude = i. Costliest element to change later

GEO 0° · ISS 51.6° · sun-sync ~98°

≠ Ω (i = how much, Ω = which way)

→ $\Delta v = 2v\cdot\sin(\Delta i/2)$ · launch latitude

WORD BANK · CONCEPT CARDS · scene 3 **C5**

RAAN (Ω) = right ascension of the scene 3
ascending node
= angle from ♈ to where the orbit
crosses the equator going north
WHY a tilted plane can face any way round
the axis. Ω pins which way
GPS: 6 planes with Ω 60° apart
≠ ω (Ω in equator, ω in orbit plane)
→ J_2 drift → sun-synchronous

ARGUMENT OF PERIAPSIS (ω) = arg. of scene 3
perigee
= angle in the orbit plane from the
node to perigee
WHY says where the near end points: over
the target, or away so apogee lingers
Molniya ω ≈ 270° → apogee hangs over
the north
≠ perigee distance (ω is an angle)
→ radial burn rotates ω · J_2 drift

WORD BANK · CONCEPT CARDS · scene 3 **C6**

TRUE ANOMALY (ν) = θ scene 3
= angle from perigee to the sat, right
now
WHY five elements freeze the orbit. ν
says WHERE on it = the clock
0° perigee · 180° apogee · ISS 360°
per 92 min
≠ mean anomaly M (even with time)
→ Kepler 2: faster near perigee

PERIAPSIS / APOAPSIS = perigee / apogee scene 3
at Earth
= nearest + fastest point / farthest +
slowest point
WHY speed is purely sideways there, so
burns go there. The two distances
define a and e
GTO: 300 km at 10.15 km/s · 35,786 km
at 1.61 km/s
≠ new orbits (-gee Earth, -helion Sun)
→ prograde @ peri raises apo

WORD BANK · CONCEPT CARDS · scene 3 **C7**

VIS-VIVA EQUATION = orbital energy eqn scene 3

= $v^2 = \mu(2/r - 1/a)$: speed anywhere on any orbit

WHY orbit energy is fixed, so speed trades with distance. One line replaces every special case

GTO: r 6,671, a 24,418 → 10.15 km/s · apogee 1.61

≠ $v = \sqrt{(\mu/r)}$ (vis-viva with a = r)

→ Δv = speed after − speed before

GRAVITATIONAL PARAMETER (μ) = GM scene 3

= G × the body's mass, as one constant

WHY every orbit formula has G×M in it, and GM is known far better than G or M alone

Earth 398,600 km^3/s^2 · Sun 1.327×10^{11} · Mars 42,828

≠ $g = 9.8\ m/s^2$ ($g = \mu/r^2$ at surface)

→ vis-viva · period · escape speed

WORD BANK · CONCEPT CARDS · scene 4 **C8**

OUTGASSING = offgassing scene 4

= gas leaking out of solid materials in vacuum

WHY materials made in air hold trapped gas. Vacuum frees it and it lands on the coldest surface = your lens

TML < 1% · CVCM < 0.1% · bake-out days at 100 °C+

≠ a leak from a tank or cabin

→ contamination · clean room · MLI

SPACECRAFT CHARGING = static build-up scene 4

= surfaces collect charge; no air to bleed it off

WHY UV knocks e^- off the sunlit side (+). Plasma e^- land on the shaded side (−). The gap arcs

kV differences measured at GEO in storms

≠ radiation damage (particles in chips)

→ ESD · conductive coating · bonding

WORD BANK · CONCEPT CARDS · scene 4 + scene 5 **C9**

STEFAN–BOLTZMANN LAW = P = σεT⁴ scene 4

= power radiated ∝ temperature to the 4th power

WHY radiation is the ONLY exit for heat in vacuum. T⁴: hot radiators work, cold ones barely

\# 1 m², ε 0.9: 300 K ≈ 410 W · 350 K ≈ 760 W · 200 K ≈ 80 W

≠ conduction (needs a temp difference)

→ radiator size · emissivity · hot spot

ATOMIC OXYGEN (AO) = O · monatomic scene 5

oxygen

= single O atoms: the main gas at 200–600 km

WHY no ozone up there, so UV splits O_2. Too thin to re-pair. Sat rams the atoms at 7.5 km/s

\# eats bare Kapton tens of µm per yr · LDEF: shredded in 5.8 yr

≠ ozone O_3 (protects) · O_2 (breathe)

→ ram / wake · coatings · drag

WORD BANK · CONCEPT CARDS · scene 5 **C10**

IONOSPHERE scene 5

= charged layer of the upper air, ~60–1,000 km

WHY UV + X-rays knock e^- off atoms. Charged gas bends and delays radio

\# GPS error up to tens of m if uncorrected

≠ ozone layer · magnetosphere

→ space weather · radio · GPS

HOMOSPHERE / HETEROSPHERE = mixed / scene 5

sorted

= below ~100 km a stirred mix; above, sorted by weight

WHY weather stirs low air into one recipe. Above the turbopause each gas settles by weight

\# N_2 + O_2 → O (200–600 km) → He → H

≠ temperature layers (tropo-, strato-)

→ turbopause · AO · geocorona

WORD BANK · CONCEPT CARDS · scene 6 **C11**

DELTA-V (Δv) = the fuel bill scene 6

= total speed change a mission needs

WHY reach is limited by speed changes, not distance. Each costs propellant → budget in km/s

surface→LEO ~9.4 · LEO→GEO 3.9 · LEO→Mars orbit ~5.6

≠ speed (Δv = sum of changes)

→ rocket equation · Hohmann = min Δv

HOHMANN TRANSFER = min-energy transfer scene 6

= half-ellipse touching both orbits; 2 burns

WHY both burns are tangent, so nothing is wasted sideways. Cheapest, NOT fastest

LEO→GEO 2.42 + 1.47 km/s, 5.3 h · Mars ~3.6 + ~2, 259 d

≠ the fastest route (it's the slowest)

→ vis-viva · TOF = π√(a³/μ) · synodic

WORD BANK · CONCEPT CARDS · scene 6 **C12**

LAMBERT'S PROBLEM scene 6

= 2 places + the time between → the orbit that fits

WHY planners start from 'be at Mars on this date'. Lambert turns that into an orbit + fuel bill

Earth Nov 2026 → Mars Aug 2027 → Δv ≈ 3.7 km/s

≠ Hohmann (one special case: 180°)

→ porkchop plot = 1000s of solutions

SYNODIC PERIOD scene 6

= time for two planets to line up the same way again

WHY planets lap the Sun at different rates. This is why launch windows exist

Earth–Mars 780 d (26 months) · Venus 584 d

≠ orbital period (Mars yr = 687 d)

→ phase angle 44° · launch window

WORD BANK · CONCEPT CARDS · scene 6 **C13**

ROCKET EQUATION = Tsiolkovsky scene 6

= Δv = Isp·g₀·ln(m₀/m_f): Δv ↔ propellant fraction

WHY a rocket carries all it will throw, so fuel grows EXPONENTIALLY with Δv

\# Isp 450 s: 3.6 km/s → 56% fuel · 9 → 87% · 60 → 99.9999%

≠ '2× fuel = 2× speed' (ratio SQUARES)

→ Isp · staging · no 20-day Mars

SPECIFIC IMPULSE (Isp) = engine mpg scene 6

= push per kg of propellant, in seconds; × 9.81 = exhaust speed

WHY engines differ in what they get from each kg. Isp is the 'miles per gallon'

\# solid ~250 s · kerosene ~320 · hydrogen 450 · ion 3,000+

≠ thrust (Isp = mpg, thrust = hp)

→ rocket equation · Section 4

WORD BANK · CONCEPT CARDS · scene 6 + scene 7 **C14**

SPHERE OF INFLUENCE (SOI) scene 6

= region where a planet's gravity beats the Sun's

WHY lets you switch 'planet rules' ↔ 'Sun rules', so each leg is a simple 2-body problem

\# Earth ≈ 925,000 km · Jupiter ≈ 48 million km

≠ a wall (gravity doesn't stop there)

→ patched conics · v∞ at the edge

GRAVITY ASSIST = flyby · slingshot scene 7

= swing past a moving planet to change speed + direction

WHY planet frame: same speed, bent path. Sun frame: you take some of the planet's motion

\# Voyager 2 +10 km/s at Jupiter · New Horizons saved 3 yr

≠ free energy (planet pays 10^{-24} m/s)

→ $\sin(\delta/2) = 1/(1 + r_p \cdot v_\infty^2/\mu)$

WORD BANK · CONCEPT CARDS · scene 7 C15

HYPERBOLIC EXCESS VELOCITY = v∞ scene 7

= speed left over once you are far from the planet

WHY near a planet speed depends on depth in the well. Far away it settles to one number

\# at Mars (Hohmann) ≈ 2.6 km/s · leaving Earth ≈ 2.9

≠ escape velocity (gives v∞ = 0)

→ C3 = v∞² · capture burn · δ

CHARACTERISTIC ENERGY (C3) = launch energy scene 7

= v∞² at Earth departure: how hard you were thrown

WHY the one number a launch provider needs. Rocket performance charts are drawn against it

\# Mars ~10–15 km²/s² · Jupiter ~80 · New Horizons ~158

≠ Δv (C3 = rocket's job)

→ v∞ = √C3 · departure burn

WORD BANK · CONCEPT CARDS · scene 7 C16

REACTION WHEEL = momentum wheel · CMG scene 7

= spinning wheel inside; the s/c turns the other way

WHY nothing to push on in vacuum, so push against yourself. No fuel, very precise

\# Hubble: 4 wheels, 0.007 arcsec · ISS: 4 CMGs

≠ thrusters (expel gas, use fuel)

→ momentum dumping · disturbances

DISTURBANCE TORQUE scene 7

= tiny unwanted twist that never stops

WHY gravity gradient, sunlight, thin air, magnetism each twist a little. Nothing damps it

\# small sat ~10^{-5} N·m · ISS several N·m. Tiny but nonstop

≠ perturbation (path, not pointing)

→ reaction wheels · momentum dumping

NOTEBOOK MAP · v3

Where the concept cards live, and what it costs

Count from the LAST page so the rule works whatever your notebook's real page count is. The page numbers in brackets assume ~95 pages; check your last page number once and adjust.

p1–5 index · equations · abbrev key	p6 → p51 LECTURE SPREADS + section maps S1–S3 already used ≈ p6–24 · room left for Sections 4, 5, 6	16 pages [p52–67] CONCEPT CARDS C1–C16 NEW · tab at 44 from end	2 talk cards	12 pages ONE-LINE WORD BANK unchanged	last 14 QUESTION BANK unchanged

The page count, honestly

Block	Pages	Why
Concept cards, S1–S3	**16**	32 terms × 9–11 lines = 309 lines. Two cards fit a 22-line page; three don't.
Concept cards, S4–S7 (estimate)	**~14**	About 25–30 more core terms at the same rate (propulsion, comms, attitude, power).
One-line word bank, whole course	12 (have) · ~15 (need)	S1–S3 is 161 lines ≈ 7½ pages; four more sections won't fit in the 4½ left.
Word bank total by end of course	**~45**	The old map gave it 12.

What that means: one ~95-page notebook cannot hold 25 lectures, a 45-page word bank and a 14-page question bank. That is ~120 pages of content. The old 2-page buffer is gone and the lecture block now ends at p51, which is the end of **Section 6**.

The plan

1. **Now:** count 44 pages back from the last page, put a sticky tab there, label the next 16 pages C1–C16 in pencil (top outer corner). Copy the eight spreads into them.
2. **Nothing else moves.** Talk cards, the 12 one-line pages and the question bank stay exactly where they are. Old tabs at 30 / 28 / 14 from the end stay.
3. **Before Section 7:** get a second small notebook. It takes Section 7's lecture spreads, the S4–S7 concept cards (C17 onward) and the one-line overflow. Number its concept pages on from C17 so the "→ C__" pointers keep working.
4. **If you would rather keep every lecture in this one book:** buy the second notebook now and put C1–C16 in it instead. Same cards, same numbering. Then this notebook's map stays exactly as v2.

My recommendation is 1–3: you are reviewing Sections 1–3 this week and need the cards next to those notes today.

Index: which page each card is on (add "→ C#" to that term's line in the one-line word bank)

ARGUMENT OF PERIAPSIS (ω) **C5**
ATOMIC OXYGEN (AO) **C9**
CENTRIPETAL FORCE **C1**
CHARACTERISTIC ENERGY (C3) **C15**
CLASSICAL ORBITAL ELEMENTS **C3**
CONSTELLATION **C2**
DELTA-V (Δv) **C11**
DISTURBANCE TORQUE **C16**
ECCENTRICITY (e) **C4**
GEOSTATIONARY ORBIT (GEO) **C2**
GRAVITATIONAL PARAMETER (μ) **C7**
GRAVITY ASSIST **C14**
HOHMANN TRANSFER **C11**
HOMOSPHERE / HETEROSPHERE **C10**
HYPERBOLIC EXCESS VELOCITY **C15**
INCLINATION (i) **C4**
IONOSPHERE **C10**
LAMBERT'S PROBLEM **C12**
OUTGASSING **C8**
PERIAPSIS / APOAPSIS **C6**
RAAN (Ω) **C5**
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ROCKET EQUATION **C13**
SEMI-MAJOR AXIS (a) **C3**
SPACECRAFT CHARGING **C8**
SPECIFIC IMPULSE (Isp) **C13**
SPHERE OF INFLUENCE (SOI) **C14**
STATION-KEEPING **C1**
STEFAN–BOLTZMANN LAW **C9**
SYNODIC PERIOD **C12**
TRUE ANOMALY (ν) **C6**
VIS-VIVA EQUATION **C7**